

Yasaman Jafari

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Education

University of California San Diego, Ph.D. in Computer Science Oct. 2021 – Present

- **GPA:** 4.0/4.0
- **Advised by:** Dr. Taylor Berg-Kirkpatrick
- **M.Sc. Degree in Computer Science** acquired Dec. 2024
- **Advanced to Candidacy** Dec. 2025

University of Tehran, B.Sc. in Computer Engineering Sept. 2016 – Feb. 2021

- **GPA:** 19.14/20.0 (4.0/4.0)
- **Advised by:** Dr. Behnam Bahrak
- **B.Sc. Thesis:** Investigating the effects of Goodreads' challenges on individuals' reading habits

Research Experience

Graduate Research Assistant, University of California San Diego – California, United States (Advisor: Dr. Taylor Berg-Kirkpatrick) Jun. 2023 – Present

- **Scalable Corpus Reasoning via "Document Souping" in State Space Models:** Designed a modular procedure that enables parallel processing and caching of document representations, eliminating the need for costly re-encoding of the entire corpus for each query.
- **GENIE - Climate Co-Scientist:** Building an agentic AI framework and novel benchmarks to accelerate weather and climate science research. (In Progress)
- **RL-based Discrete Prompt Optimization:** Developed multi-objective optimization techniques for RL-based prompt optimization, improving balance across competing rewards in NLP tasks like style transfer and translation.

Graduate Research Assistant, University of California San Diego – California, United States (Advisor: Dr. Babak Salimi) Oct. 2021 – Jun. 2023

- Repaired datasets using optimal transport to enforce conditional independence constraints and improve robustness
- Quantified subgroup heterogeneity and intersectional effects to mitigate performance disparities and improve calibration

Undergraduate (Volunteer) Research Assistant, University of Tehran Science and Technology Park – Tehran, Iran (Advisor: Dr. Behnam Bahrak) Oct. 2019 – Feb. 2021

- Analyzed GoodReads challenges and the potential effects of publicly announcing goals

Research Papers

Black-Box Forensics for Conversational LLM Agents 2026

I. White, Y. Jafari, T. Berg-Kirkpatrick

[Under Submission]

Studying the Soupability of Documents in State Space Models 2025

Y. Jafari, Z. Wang, L. Bergen, T. Berg-Kirkpatrick

[COLM 2026]

Zephyrus: An Agentic Framework for Weather Science 2025

S. Varambally, M. Fisher, J. Thakker, Y. Chen, Z. Xia, Y. Jafari, R. Niu, M. Jain, V. V. Manivannan, Z. Novack, L. Han, S. Eranky, S. R. Cachay, T. Berg-Kirkpatrick, D. Watson-Parris, Y. Ma, R. Yu

[ICLR 2026]

Zephyrus: An Agentic Framework for Weather Science 2025

S. Varambally, M. Fisher, J. Thakker, Y. Chen, Z. Xia, R. Niu, Y. Jafari, V. V. Manivannan, Z. Novack, L. Han, S. Eranky, S. R. Cachay, T. Berg-Kirkpatrick, D. Watson-Parris, Y. Ma, R. Yu

[NeurIPS 2025 Workshop on Scaling Environments for Agents (SEA)]

Aquilon: Towards Building Multimodal Weather LLMs S. Varambally, V. Manivannan, Y. Jafari , L. Han, Z. Novack, Z. Xia, S. Rühling Cachay, S. Eranky, R. Niu, T. Berg-Kirkpatrick, D. Watson-Parris, Y. Ma, R. Yu <i>[ICML 2025 Workshop on Assessing World Models]</i>	2025
ClimaQA: An Automated Evaluation Framework for Climate Foundation Models V. Manivannan, Y. Jafari , S. Eranky, S. Ho, R. Yu, D. Watson-Parris, Y. Ma, L. Bergen, T. Berg-Kirkpatrick <i>[ICLR 2025] arXiv:2410.16701</i>	2024
MORL-Prompt: An Empirical Analysis of Multi-Objective Reinforcement Learning for Discrete Prompt Optimization Y. Jafari , D. Mekala, R. Yu, T. Berg-Kirkpatrick <i>[EMNLP Findings 2024] arXiv:2402.11711</i>	2024
Investigating the effects of Goodreads challenges on individuals reading habits Y. Jafari , N. Sabri, B. Bahrak <i>[Pre-print] arXiv:2012.03932</i>	2020

Work/Internship Experience

Siri SW Intern , Apple Inc. – Cupertino, United States • Research on State Space Models for Agentic models	Jun. - Sept. 2026
Software Development Engineer 2 , Populus Group (On Assignment for Microsoft Research) – Part-time, Remote (Continuation of Summer 2025 MSR Internship work) • AI for Science and Agentic Frameworks	Feb. - Jun. 2026
Research Intern , Microsoft Research – Redmond, United States • Developed an agentic framework for power grid planning and risk management from hurricanes and other extreme weather events	Jun. – Sept. 2025
Data Science IntBlack-Box Forensics for Conversational LLM Agent (KYC - Adverse Media), Moody's Analytics (Moody's) – New York, United States (Remote) • Created evaluation data for an existing model used in Adverse Media Highlighting • Evaluated the model and conducted a thorough analysis of its performance, identifying areas of failure, and proposing various options for improving the model	Jun. - Aug. 2023
Data Analysis and Modeling Intern , Moody's Analytics (Moody's) – United States (Remote) • Mitigated the run-time bottlenecks of working with large data using Spark • Converted estimation codes from R to Python for efficiency purposes	Jun. - Aug. 2022
Software Engineering Intern , Supertext AG – Zurich, Switzerland • Implemented various front-end and back-end features such as Instant Translation and SSO authentication	Feb. - Jul. 2020

Teaching Experience

Graduate Teaching Assistant , University of California San Diego – California, United States • CSE 251A: Machine Learning: Learning Algorithms, Dr. Taylor Berg-Kirkpatrick • CSE 158: Web Mining and Recommender Systems, Dr. Julian McAuley • CSE 256: Statistical Natural Language Processing, Dr. Ndapa Nakashole • CSE 250A: Probabilistic Reason and Learning Course, Dr. Taylor Berg-Kirkpatrick	Jan. – Mar. 2026 Sept. – Dec. 2023 Mar. – Jun. 2023 Sept. - Dec. 2022
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Honors and Awards

• Google CS Research Mentorship Program (CSRMP) Scholar • Fellowship Award for Graduate Study at UC San Diego • Finalist for the Best Undergraduate Project Award • 3rd rank (among 88 students) - FOE Top Students Award • 1st rank (among 88 students) - FOE Top Students Award	May. 2023 Oct. 2021 - Jun. 2022 Feb. 2021 Jun. 2018 Jun. 2017
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Services

Conference Reviewer: COLM 2026, ICLR 2026, NeurIPS 2025, ICML 2025, NeurIPS 2024, ICML 2024, ICLR 2024, NeurIPS 2023, ICML 2023, FAccT 2023

UCSD Graduate Women in Computing (GradWIC) Mentor Served as a mentor in the GradWIC mentorship program for incoming graduate students from underrepresented backgrounds (Academic year 2023-2024)

Conference Volunteer Student ICDE 2023

Skills

Programming Languages: Python, C++, C, C#, Java, JavaScript, Ruby

Languages: Persian (Native Language), English (Professional Working Proficiency – TOEFL 119/120), German (Intermediate – B1)